Week 1 Notes

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Contents

1 What is a Database?

1.1 Databases

Simply, **database** is a collection of information.

Fields The individual kinds of information are organized into **columns**.

Records Each set of information is organized into **rows**.

Table Fields and records make up a **table**.

Datbase One or more tables make up a **database**.

> The rules that regulate rules and layouts of the database are called the "*schema*" of the database.

1.2 Databases vs. Spreadsheets

While both databases and spreadsheets have fields and records:

- Database tables can have specific relationships to each other.
- Databases allow us to manage data and ask questions about data.

2 Why Use a Database

Datbases let us work with large amounts of data easily. They offer security features to control access to information, and help to reduce redunancy.

3 The Client-Sever Model

All relational databases operate on the **client-server model**. This means that there's two components that work together to make a complete system.

Database Sever Performs all relational database management tasks. This is where data is structured, stored, and secured.

Interface Client Whether it's graphical or text-based, it allows users to connect and control activities on the sever.

3.1 Benefits of the Client-Server Model

This system excels at allowing a tailored-system to suit particular needs.

1. Database servers can accept connections from many simultaneous clients.
2. A single client can connect to multiple servers.
3. Most RDBMSs come with preffered client direct from the same vendor.
4. Third-party clients are often available though.

4 Relational Database Management Systems

Relational Database Management Systems, or **RDMBS** for short, gives you the tools that you need in order to build database to meet your unique set of requirements.

4.1 Data Structures

- Create tables
- Define column parameters
- Formalize relationships
- Create data validation rules

4.2 Data Storage

- Organize values in tables
- Build and search indexes
- Maintain redundant data
- Regular backups

4.3 Data Security

- Control access
- Manage user login
- Keeps logs for auditing

These are three main responsibilties for an **RDBMS**.

4.4 DB Scope

- Desktop
- Server
- Cloud

Depending on your needs, you can choose the scope you want your database to cover.

5 Storing Data Efficiently

By splitting data efficiently, you can minimize the amount of space you need while maximizing the granularity in your organization.

6 Relational DB

A specific type of database where data is:

- Structured into tables. Rows/records and columns/fields.
- With explicitly defined relationships between tables.

Keys consisting of one or more fields, are used to identify records.

- A primary key uniquely identifies a record
- a foreign key defines a relationship between two tables by referring to a record in a different table

7 Structured Query Language

- The programming language of relational databases.
- Developed in the 1970's
- Interface to using the features provided by the RDBMS

  DML Data Manipulation Language (focus for semester)

  DDL Data Definition Language

  DCL Data Control Language